TimeClaw: Design a mark for TimeClaw

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Abstract

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1 Introduction

2 Introduction

Time series data serves as the core medium for capturing evolutionary patterns across complex cyber-physical systems, including smart cities, financial networks, climate monitoring, and macroeconomic analysis [1**?**]. Accurate decoding and forward-looking forecasting of these massive temporal signals underpin digital twins, automated decision engines, and early-warning systems for extreme events [**? ?**]. Traditional analytical pipelines rely on domain experts for feature engineering, model selection, and workflow orchestration—a process that incurs high labor costs and struggles to cope with ubiquitous distribution shifts and variable heterogeneity in real-world deployments [2, 3]. Recently, Large Language Models (LLMs) have opened the door to a paradigm shift: by combining time series analysis with the reasoning capabilities of LLMs, researchers are building *agentic time series analysis* systems—"AI data scientists" that can autonomously invoke external tools and execute multi-step logical deduction [4**?** , 5].

Despite the promise of this paradigm, existing methods suffer from three structural limitations on the path toward general and reliable temporal intelligence:

(i) **Numerical hallucination and opaque reasoning.** Foundation models that map temporal values to discrete tokens for autoregressive generation (e.g., Chronos [6], TimesFM [7]) are susceptible to numerical hallucinations and lack the capacity for multi-step logical deduction in complex physical scenarios with exogenous variables [8**? ?**].

(ii) **Static tool orchestration without autonomous exploration.** Agent systems based on static tool augmentation (e.g., TS-Agent [5], TSAD-Agents [9]) solidify the analytical process into a fixed, unidirectional pipeline. They lack the ability to autonomously design experimental schemes and dynamically explore alternative paths when confronted with high-entropy or uncertain environments.

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(iii) **Frozen knowledge without incremental learning.** Systems trained via reinforcement learning or behavior cloning (e.g., TimeART [4]) package execution experiences into opaque network parameters in a single pass. This renders incremental updates and explicit self-correction impossible: the knowledge is locked to a specific checkpoint and cannot be inspected, edited, or transferred [**?** **?**].

These rigidities collectively precipitate a critical vulnerability: when confronting out-of-distribution shifts or unforeseen anomalies, current agents lack the evolutionary capacity of human experts—the ability to extract causal relationships via trial-and-error and achieve incremental self-correction. In mission-critical domains such as power-grid monitoring and financial risk control, this inability to accumulate experience introduces unacceptable risks of uninterpretable failures [**?** 2**?**].

To overcome these bottlenecks, we propose **TimeClaw**, a native multimodal time series autonomous agent platform. The core innovation lies in a natural-language-based *learnable functional module* that elevates the system from a rigid pipeline to a self-evolving architecture. TimeClaw introduces four tightly coupled components, each targeting a specific deficiency:

(i) **Experience-Augmented Executor** (§4.3). To address numerical hallucinations and opaque reasoning, TimeClaw adopts a native multimodal fusion paradigm within a single ReAct agent. Heavy numerical computations are offloaded to an accurate external tool library; the LLM conducts logical reasoning exclusively on factual tool outputs. Five layers of persistent knowledge are injected into the agent's prompt, enabling tool-selection decisions informed by accumulated experience.

(ii) **Contrastive Parallel Exploration** (§4.4). To overcome the rigidity of static pipelines, the agent dynamically spawns multiple sub-agents to conduct controlled experiments across different hypotheses and tool compositions when facing analytical bottlenecks. This yields high-quality, quantitative comparative evidence through a process akin to automated ablation studies.

(iii) **Hierarchical Knowledge Distillation** (§4.5). To break the limitations of knowledge freezing, a multi-tier distillation cascade progressively refines raw trial-and-error experiences into human-readable, editable causal rules and standard operating procedures, while automatically resolving conflicts with outdated knowledge. All learned knowledge is maintained as text rather than model weights.

(iv) **Closed-Loop Integration.** Distilled knowledge is continuously injected back into the Executor's prompt, completing the cycle *Execute* $\rightarrow$ *Explore* $\rightarrow$ *Distill* $\rightarrow$ *Inject* $\rightarrow$ *Execute* and enabling cumulative capability evolution across training episodes.

The main contributions of this paper are:

- **C1 (Architecture):** We propose TimeClaw, a general time series analysis agent framework that seamlessly integrates multimodal native reasoning, exploratory trial-and-error, and incremental experience learning into a fully operational closed loop, overcoming the fragmentation between execution and learning in existing systems.
- **C2 (Methodology):** We introduce a dual-engine learning paradigm combining contrastive parallel exploration with natural-language-based hierarchical distillation. By autonomously conducting micro-experiments for causal attribution and condensing insights into text-based rules, this paradigm enables continuous capability evolution without modifying underlying model parameters.
- **C3 (Experiments):** Extensive experiments across diverse real-world time series tasks on the MTBench benchmark demonstrate that TimeClaw achieves state-of-the-art performance, significantly outperforming existing parameter-frozen agentic systems while offering unprecedented transparency and interpretability.

3 Related Work

LLMs for time series analysis. Deep learning has driven substantial progress in time series analysis, with Transformer-based architectures emerging as the dominant backbone for tasks including forecasting, anomaly detection, and classification [1, 3, 15, 2]. Building on this, researchers have developed time series foundation models for zero-shot generalization: autoregressive tokenization approaches such as Chronos [6] and TimesFM [7] map numerical values to discrete tokens for

Table 1: Comparison of TimeClaw with representative time series analysis systems on five agentic dimensions: ① Agentic Orchestration, ② Multimodal Reasoning, ③ Contrastive Exploration, ④ NL-Experience Learning, ⑤ Hierarchical Distillation. ✓ full support, △ partial, — none.

System	Year	①	②	③	④	⑤
MAS4TS [10]	2026	△	✓	—	—	—
TimeART [4]	2026	✓	△	—	—	△
TSAD-Agents [9]	2026	△	✓	—	—	—
TS-Debate [11]	2026	—	✓	△	—	—
Comp.-Enh. MAS [12]	2025	△	—	✓	△	△
TS-Agent [5]	2025	✓	—	—	—	—
TimeXL [13]	2025	—	△	—	—	△
From News to Forecast [14]	2024	△	△	—	—	△
TimeClaw (Ours)	2026	✓	✓	✓	✓	✓

universal forecasting, while masked pre-training variants like MOMENT [16] and Moirai [17] adopt encoder-centric designs to learn transferable temporal representations. A complementary line of research asks whether LLMs can serve as general-purpose reasoners for time series. Reprogramming-based methods such as Time-LLM [18] and CALF [19] project patch embeddings into the LLM token space to exploit frozen language priors for forecasting. Multimodal fusion approaches Chat-Time [20] and TimeVLM [21] jointly encode numerical signals and textual context for cross-modal question answering. Encoder-augmented architectures [22] further demonstrate that native continuous representations substantially outperform text serialization for perception and deductive reasoning. Dedicated TSQA benchmarks have catalyzed this direction: MTBench [23] provides paired financial and weather series with news narratives; ChatTS [24] constructs synthetic corpora aligning time series with natural language; and Time-MQA [25] spans forecasting, anomaly detection, and open-ended reasoning across twelve domains. Despite this progress, benchmarking studies [26, 8] consistently reveal that raw LLMs suffer from *numerical hallucination* when processing serialized values, and their quantitative capabilities remain far below specialized analytical tools.

Agentic Systems for time series analysis. The recognition that LLMs are weak numerical computers but strong reasoners has motivated agentic time series systems that delegate computation to external tools. TS-Agent [5] and TSAD-Agents [9] couple an LLM with fixed analytical libraries and execute a pre-defined unidirectional pipeline for anomaly detection and forecasting. TimeART [4] advances this paradigm by constructing a 100k ReAct-style [27] expert trajectory corpus and training an 8B reasoning model through a four-stage curriculum covering world-model warm-up, behavior cloning, and self-reflection. Multi-agent variants (MAS4TS [10], MERIT [28]) decompose complex analytical tasks across specialized sub-agents to improve coverage and reduce single-point errors. Despite these advances, three structural limitations persist. First, all existing agent systems rely on *static tool orchestration*: the analytical pipeline is either hard-coded or trained offline, with no mechanism to explore alternative hypotheses at inference time when initial tool choices fail. Second, systems trained via behavior cloning or sparse-reward reinforcement learning package execution experience into opaque network weights, making knowledge inspection, selective editing, and transfer to new distributions impossible. Third, none implements a closed loop where experience continuously accumulates and is injected back to improve future executions. Enabling agents to learn from their own trial-and-error—rather than pre-collected demonstrations alone—is an active frontier: experience replay and self-reflection methods [29, 30] allow agents to annotate trajectories with natural-language rationales, while text-based knowledge libraries [30, 31] demonstrate that editable symbolic representations are more transparent and updateable than weight-encoded memories. TimeClaw addresses all three limitations simultaneously: Contrastive Parallel Exploration dynamically spawns competing sub-agents to generate high-quality comparative evidence; Hierarchical Knowledge Distillation progressively refines raw experience into editable causal rules; and the closed-loop design continuously injects distilled knowledge back into the executor without any parameter update. As shown in Table 1, existing systems address at most two of the five dimensions simultaneously. For instance, TimeART achieves strong tool orchestration and explores model internalization through supervised fine-tuning, but lacks controlled experimentation (③) and human-readable incremental learning (④). TS-Debate demonstrates multimodal reasoning through inter-agent debate, yet no knowledge persists across

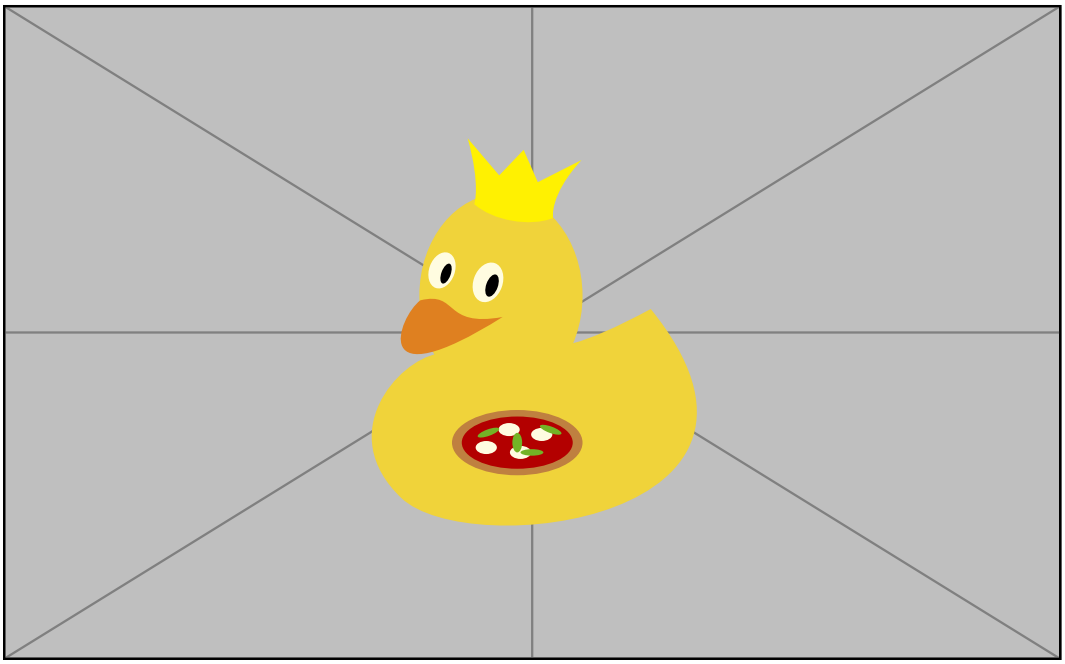


Figure 1: Overview of the TimeClaw framework. The Experience-Augmented Executor (§4.3) performs multimodal reasoning via a memory-injected ReAct loop. In training mode, Contrastive Parallel Exploration (§4.4) spawns sub-agents for controlled experimentation, and results are progressively refined through Hierarchical Knowledge Distillation (§4.5). Distilled knowledge is injected back into the Executor's prompt, closing the self-evolution loop.

instances. By integrating a learnable natural-language module, TimeClaw is the first framework that simultaneously satisfies all five dimensions within a single closed loop.

4 Methodology

4.1 Problem Setting

We study *multimodal time-series reasoning* (TSR). Each instance consists of a numerical time series $\mathbf{x}_{1:T} \in \mathbb{R}^{T \times d}$ with T time steps and d channels, a natural-language task specification q (e.g., "forecast the next 7 days" or "classify the trend direction"), optional textual context $\mathcal{C}$ (e.g., financial news or weather reports), and optional visual data $\mathcal{V}$ (e.g., charts or sensor images). The goal is to produce an answer $\hat{a} \in \mathcal{A}(q)$, where the answer space is induced by the task—a numerical vector for forecasting, a categorical label for classification, or a natural-language response for question answering.

We assume access to a pool of K external tools $\mathcal{T} = \{T_1, \ldots, T_K\}$, each implementing an atomic analytical function (e.g., ARIMA forecasting, trend detection, STL decomposition). Tools are black-box: the agent observes only their input schemas and output values. We further assume a labeled training set $\mathcal{D}_{\text{train}} = \{(\mathbf{x}^{(i)}, q^{(i)}, a^{(i)})\}_{i=1}^{M}$ for experience accumulation; at test time, no ground truth is available.

The central challenge is that no single tool or fixed pipeline performs best across all data conditions and task types. The agent must **(1)** *autonomously compose* multi-step tool chains conditioned on data characteristics, **(2)** *learn from experience* which compositions work under which conditions, and **(3)** *persist and reuse* this knowledge across tasks without modifying the underlying LLM weights.

4.2 Framework Overview

TimeClaw addresses this challenge through four tightly coupled components (Figure 1):

Table 2: Five-layer memory injection in the Executor prompt. Each layer occupies a distinct position in the LLM context, ensuring that knowledge of different abstraction levels reaches the agent at the appropriate stage of its reasoning.

Layer	Position	Content	Function
SOUL	System prompt	Fixed behavioral principles	Stable reasoning identity
MEMORY	System prompt	Causal decision rules (§4.5.2)	High-level strategy
DAILY NOTES	User message	Retrieved raw experiences (§4.5.1)	Concrete precedents
SKILLS	User message	Operational SOPs (§4.5.4)	Step-by-step procedures
TOOLS.MD	Tool descriptions	Per-tool usage notes (§4.5.3)	Empirical tool guidance

1. **Experience-Augmented Executor** (§4.3): A ReAct-style agent whose prompt is augmented with five layers of persistent knowledge, enabling tool-selection decisions informed by prior experience.
2. **Contrastive Parallel Exploration** (§4.4): A training-time mechanism in which the agent autonomously spawns parallel sub-agents to try different tool compositions, evaluates all outcomes against ground truth, and produces a structured comparative analysis.
3. **Hierarchical Knowledge Distillation** (§4.5): A multi-tier memory system that transforms raw exploration outcomes into increasingly abstract knowledge artifacts—from execution logs to causal rules to operational procedures—with automatic conflict resolution.
4. **Closed-Loop Integration**: Distilled knowledge is injected back into the Executor's prompt, closing the loop so that each training episode improves subsequent executions.

These components form the cycle *Execute* → *Explore* → *Distill* → *Inject* → *Execute*. In test mode only the first and last stages are active; in train mode all four stages run, enabling the system to accumulate transferable knowledge.

4.3 Experience-Augmented Executor

ReAct with memory injection. The Executor implements a Think→Act→Observe loop [**?**]. At each step $t \in \{1, \ldots, S_{\max}\}$, the LLM receives the conversation history $h_{1:t-1}$ and produces either a tool call (T_j, args_t) or a terminal answer $\hat{a}$:

$$(\text{thought}_t,\ \text{action}_t)\ \sim\ p_\theta\big(\cdot \mid h_{1:t-1},\ \mathcal{P}(\mathcal{M})\big), \tag{1}$$

where $\mathcal{P}(\mathcal{M})$ is the *memory-augmented prompt* constructed from the hierarchical memory $\mathcal{M}$. If the action is a tool call, the tool executes and returns observation $o_t = T_j(\mathbf{x}, \text{args}_t)$, which is appended to h. The loop terminates when the agent declares completion or $S_{\max}$ is reached.

The key departure from standard ReAct is $\mathcal{P}(\mathcal{M})$, which injects five layers of knowledge at strategically different positions in the context (Table 2). Abstract rules (MEMORY) in the system prompt bias the agent's overall strategy; concrete notes and SOPs in the user message inform immediate decisions; tool-level annotations are placed alongside tool descriptions where they are most actionable.

Multimodal reasoning within a single agent. Rather than assigning separate agents to different modalities [**?**], TimeClaw processes all modalities within a single ReAct loop. During the perception phase, the input is serialized into a unified context: the numerical array $\mathbf{x}_{1:T}$ (or head/tail excerpts for long series), full-text documents from $\mathcal{C}$, and image descriptors from $\mathcal{V}$. The agent decides autonomously which modality to analyze at each step and can cross-reference findings across modalities—for example, using text-derived sentiment to disambiguate a visually ambiguous trend. This avoids the $O(R \cdot |\mathcal{M}|)$ LLM-call overhead of multi-agent debate while preserving cross-modal reasoning.

Preprocessing. Before the agent begins reasoning, a deterministic preprocessing step (no LLM calls) extracts a data fingerprint $\mathcal{F} = \langle \text{length}, \text{channels}, \text{stationarity}, \text{trend}, \ldots \rangle$ by running a fixed suite of profiling tools on $\mathbf{x}_{1:T}$. This fingerprint provides the agent with a structured data summary and seeds the semantic retrieval query for memory lookup (§4.5.1). A fingerprint-based strategy recommender suggests an initial approach, but the agent is free to override this recommendation.

4.4 Contrastive Parallel Exploration

A central design principle of TimeClaw is that the agent learns not from external error diagnosis, but from its own structured self-experimentation. In training mode, two additional mechanisms are activated.

Sub-agent spawning. The agent can invoke a built-in `spawn_subagent` action that forks up to N independent sub-agents. The main agent specifies a set of approach descriptions $\{d_i\}_{i=1}^{N}$; each sub-agent Sub_i inherits the full memory $\mathcal{M}$ and the parent's conversation history, but executes an independent mini-ReAct loop of up to $S'_{\max}$ steps:

$$\hat{a}_i,\ \log_i\ =\ \mathrm{Sub}_i\big(d_i,\ \mathbf{x},\ \mathcal{M},\ h_{\mathrm{parent}}\big)\quad \forall\, i \in \{1,\ldots,N\}. \tag{2}$$

Sub-agents execute concurrently and return predictions, tool chains, and reasoning traces. This design deliberately encourages *diversity*: different sub-agents may use different forecasting models, apply different preprocessing steps, or adopt entirely different strategies (e.g., statistical analysis vs. LLM-based reasoning).

Quantitative evaluation and ground-truth isolation. After collecting sub-agent results, the main agent evaluates each prediction against the ground truth a^* using a training-only evaluation tool:

$$m_i = \mathrm{Eval}(\hat{a}_i,\ a^*,\ \tau)\quad \forall\, i \in \{1,\ldots,N\}, \tag{3}$$

where τ is the task type and m_i is a task-specific metrics dictionary. The agent then performs comparative analysis: it examines $\{(m_i, \log_i)\}_{i=1}^{N}$ and produces a *causal attribution*—not merely "approach C scored best," but *why* (e.g., "pre-decomposition removed the seasonal component that caused forecaster overshoot").

A critical design concern is preventing ground-truth leakage into test-time behavior. Two safeguards are applied: **(i)** after the agent analyzes the evaluation output in the current reasoning step, raw ground-truth values are automatically stripped from the conversation history before the next step—the agent retains only the computed metrics and its own qualitative analysis; **(ii)** all distillation prompts (§4.5) explicitly reject rules that reference ground-truth distributions or evaluation-only tools.

Learning summary. The training episode concludes with a structured learning summary containing an *insight* (causal explanation of what worked and why) and a *recommendation* (actionable strategy for similar future tasks). This summary, together with the full tool chain and all evaluation metrics, constitutes the raw material for knowledge distillation.

Relation to existing learning paradigms. The contrastive exploration mechanism can be viewed as the agent autonomously designing and executing controlled experiments: it holds the data constant, varies the tool composition, measures outcomes, and attributes causes. This is qualitatively different from error-diagnosis approaches where an external teacher analyzes a student's failures post-hoc [**?**]. In TimeClaw, the agent initiates, executes, and interprets its own experiments, requiring no stronger teacher model.

4.5 Hierarchical Knowledge Distillation

The learning summaries produced in §4.4 are progressively refined into persistent knowledge through a multi-tier distillation cascade. Each tier increases in abstraction and decreases in volume, balancing specificity with generalizability (Figure **??**).

4.5.1 Daily Notes: Raw Experience Store

Each training episode produces a structured note recording the outcome (success/failure), the tool chain used, all evaluation metrics, and the agent's qualitative analysis. Notes are stored in an append-only log and indexed in a vector store for semantic retrieval. During inference, the top-k notes most relevant to the current task are retrieved and injected into the prompt.

Before distillation, a **conflict resolution** step identifies outdated or contradicted notes. An LLM examines related notes retrieved for the same context and marks those superseded by newer evidence for removal. This prevents stale experience from corrupting downstream distillation.

4.5.2 MEMORY: Causal Decision Rules

When the note count for a given scope (domain, task) exceeds a threshold δ_1, a distillation step synthesizes *causal decision rules* from the accumulated notes. Each rule follows the template:

$$[\,\textsc{When}\ c\,]\ \rightarrow\ [\,\textsc{Do}\ s\,]\ \ (\textsc{Because}\ r;\ \textsc{Evidence:}\ e\,) \tag{4}$$

where c is a data condition, s is a tool-selection strategy, r is a causal reason, and e is quantitative evidence from training runs. The distillation prompt enforces several constraints: each rule must specify both a data condition and a tool recommendation; rules referencing ground-truth distributions or evaluation-only tools are rejected; and the total number of rules per scope is bounded to prevent prompt bloat.

Cross-domain synthesis. When the total MEMORY entries across all scopes reach δ_2, a second distillation step synthesizes universal insights that apply across domains, referencing tool *categories* (e.g., "foundation models" vs. "statistical methods") rather than specific tool names.

4.5.3 TOOLS.md: Per-Tool Usage Notes

A parallel distillation path computes aggregate statistics per tool—success rate, average metrics, and failure contexts—and synthesizes a concise usage note for each tool. These notes are appended to the corresponding tool descriptions in the prompt. Tools with fewer than a minimum number of recorded uses are annotated as "Insufficient data," which the training prompt interprets as a signal to prioritize exploring those tools in future runs, creating a natural exploration incentive.

4.5.4 SKILLS: Standard Operating Procedures

From the MEMORY rules for each scope, a final distillation step generates a Standard Operating Procedure (SOP) containing a numbered procedure, common pitfalls, and tool recommendations. This SOP is injected into the user message at inference time, providing the agent with a complete playbook for the given (domain, task) pair.

4.5.5 Distillation Cascade and Self-Maintenance

The full distillation chain is triggered automatically after each new note:

$$\textsc{Notes} \xrightarrow{\delta_1} \textsc{Memory} \xrightarrow{\delta_2} \textsc{Cross-Domain} \tag{5}$$

$$\textsc{Notes} \xrightarrow{\delta_1} \textsc{Tools.md} \tag{6}$$

$$\textsc{Memory} \xrightarrow{\text{on update}} \textsc{Skills} \tag{7}$$

Crucially, each distillation step is not merely additive: the LLM reviews *all* existing rules against the new evidence, merges overlapping rules, and removes those contradicted by newer data. This self-maintenance property ensures that the knowledge base does not accumulate stale or conflicting entries over time—in contrast to flat skill libraries [**?** **?**] where knowledge artifacts are generated once and remain static.

4.6 Discussion: Design Choices

We briefly justify three key design decisions.

Single agent vs. multi-agent debate. Multi-agent debate frameworks [**?**] achieve strong single-shot performance by dedicating separate agents to each modality. However, they require $O(R \cdot |\mathcal{M}|)$ LLM calls per instance (where R is the number of debate rounds and $|\mathcal{M}|$ is the number of modalities), and no knowledge persists across instances. TimeClaw uses a single agent with accumulated experience, reducing per-instance cost while enabling improvement over time. We quantify this trade-off in §**??**.

In-context knowledge vs. parameter updates. Tool-augmented approaches such as TimeART [4] fine-tune model weights on expert trajectories. This produces *model-locked expertise*: the knowledge is entangled with a specific checkpoint and cannot be inspected, edited, or transferred to a successor model. TimeClaw externalizes all learned knowledge as human-readable text, enabling version control, auditing, and zero-cost migration to new LLM backends.

Agent-initiated vs. teacher-diagnosed learning. Concurrent work [**?**] generates knowledge artifacts via external teacher-model diagnosis of student failures. TimeClaw instead has the agent design and execute its own comparative experiments via sub-agent spawning. This produces richer learning signal—not only "what failed and why," but "which of several valid approaches is best and what data characteristics explain the difference"—and requires no stronger teacher model. We ablate the contribution of contrastive exploration vs. single-approach learning in §**??**.

5 Experiments

6 Conclusion

Please read the instructions below carefully and follow them faithfully.

6.1 Style

Papers to be submitted to NeurIPS 2026 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. **Papers that exceed the page limit will not be reviewed (or in any other way considered) for presentation at the conference.** Additional pages *containing acknowledgments, references, checklist, and optional technical appendices* do not count as content pages.

The margins in 2026 are the same as those in previous years.

Authors are required to use the NeurIPS LaTeX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for desk rejection.

6.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

https://neurips.cc.

The only supported style file for NeurIPS 2026 is `neurips_2026.sty`, rewritten for LaTeX 2_ε. **Previous style files for LaTeX 2.09, Microsoft Word, and RTF are no longer supported.**

The LaTeX style file contains three optional arguments:

- `final`, which creates a camera-ready copy,
- `preprint`, which creates a preprint for submission to, e.g., arXiv,
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Preprint option If you wish to post a preprint of your work online, e.g., on arXiv, using the NeurIPS style, please use the `preprint` option. This will create a nonanonymized version of your work with the text "Preprint. Work in progress." in the footer. This version may be distributed as you see fit, as long as you do not say which conference it was submitted to. Please **do not** use the `final` option, which should **only** be used for papers accepted to NeurIPS.

At submission time, please omit the `final` and `preprint` options. This will anonymize your submission and add line numbers to aid review. Please do *not* refer to these line numbers in your paper as they will be removed during generation of camera-ready copies.

The file `neurips_2026.tex` may be used as a "shell" for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in Sections 7, 8, and 9 below.

7 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow ¼ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 9 regarding figures, tables, acknowledgments, and references.

8 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

8.1 Headings: second level

Second-level headings should be in 10-point type.

8.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

9 Citations, figures, tables, references

These instructions apply to everyone.

9.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

produces

Hasselmo, et al. (1995) investigated…

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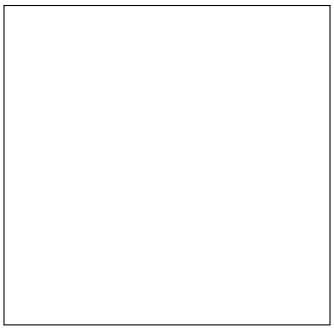


Figure 2: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

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```

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9.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number[1] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.[2]

9.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for the first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

9.4 Tables

All tables must be centered, neat, clean, and legible. The table number and title always appear before the table. See Table 3.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for the first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

https://www.ctan.org/pkg/booktabs

[1] Sample of the first footnote.

[2] As in this example.

Table 3: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\approx$100
Axon	Output terminal	$\approx$10
Soma	Cell body	up to 10^6

This package was used to typeset Table 3.

9.5 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using $$ anyway; see https://tex.stackexchange.com/questions/503/why-is-preferable-to and https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath for more information.)

9.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes. Please note that pages should be numbered.

10 Preparing PDF files

Please prepare submission files with paper size "US Letter," and not, for example, "A4."

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF files uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdffonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

  Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

10.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command when necessary.

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A Technical appendices and supplementary material

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